Optimization & Operations Research Engineer

(MIP | Routing | Scheduling | Heuristics)

Lahari Sengupta, PhD

Who am I?

- Need **business** decisions?
  - I help you to take **best decisions** by implementing **mathematical optimization models** for the **real-world problems** using Mixed Integer Programming (MIP), heuristics, and metaheuristics

- **My expertise**
  - **Scheduling** and **resource allocation** problems
  - **Vehicle routing** problems
  - **Supply chain** optimization
  - Commercial (**Gurobi, CPLEX, Fico Xpress**) or open-source (**OR-tools, Pyomo, Optaplanner**) solvers

- **Why me?**
  - **PhD** in Optimization
  - **Industry** experience
  - Clean, robust, **production-ready** solution

Industry work sample

1. Freight routing

Optimizing Profitable Freight Routes for a Mid-Mile Marketplace

Stabilizing a circular pickup-delivery bundling engine for a European freight marketplace : A real-world Vehicle Routing Problem (VRP) scenario

9-hrs
Shift Limit

X euro/km
Profitability threshold

Closed
Tour routing

Optimizing Profitable Freight Routes for a Mid-Mile Marketplace

The Pain Points

- Empty return trips eating into driver margins
- Manual quote negotiations — slow & unreliable
- Unpredictable per-route profitability
- Low vehicle utilization on LTL shipments
- Difficulty staying within 9-hour shift windows

The Goal

Automatically bundle multiple LTL shipments into profitable, fully-utilized circular routes — without manual intervention.

Revenue Constraint:

```
∑ OrderValue  ≥  X × RouteDistance
```

- ✓ Respect pickup-before-delivery precedence
- ✓ Stay within 9-hour driving shift
- ✓ Close tour back to origin pickup
- ✓ Filter insertions: Δvalue $\geq 0.60 \times \Delta$distance

Optimizing Profitable Freight Routes for a Mid-Mile Marketplace

Tech-stack used

- Open street routing machine (OSRM) distances
- Google OR tools
- Python

Developed Solution

Distance matrix
- Compute distance matrix for all to all pickup-delivery

Algorithm
A. Cheapest insertion
B. Tabu search

Route scoring
- Each candidate route ranked by cost/km. Only routes above threshold kept

Feasibility check
- Shift timing validation, route profitability, and delivery only after pickup are respected

Business impact

✓ **Real-world** OSRM distances, not straight-line estimates

✓ **Revenue-feasibility** built into the routing core

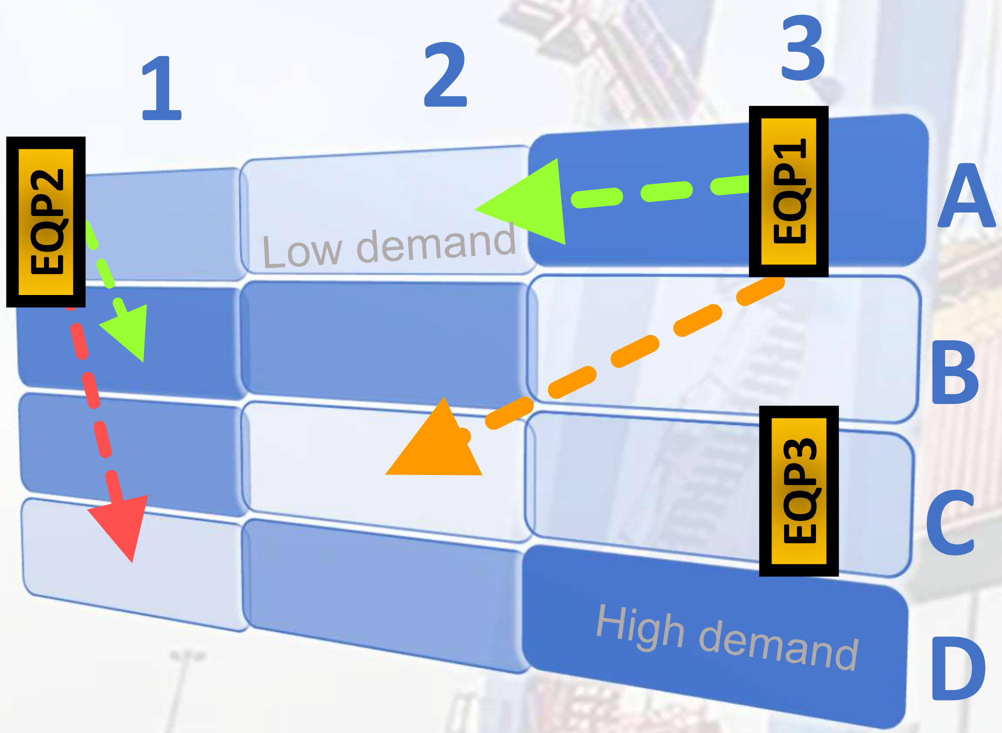
✓ Marketplace-native: elastic, **assignmentless** fleet

✓ Audited, **stabilized**, and roadmapped for scale

2. Equipment scheduling

Equipment scheduling in container terminals

Operational Cost | **Equipment Capacity** | **Throughput in a yard**



Yard structure — Horizontal lanes – vertical blocks – like a grid based yard structure

Spatio-temporal demand — Specific workload demand per block, per time period. Demand varies throughout the operational horizon

Equipment constraints — Equipment has specific capacity limits and fixed starting locations. Movement requires time

Dynamic movement costs — Moving between blocks incurs cost proportional to distance. Closer moves are cheaper; inter-lanes moves are costly

Equipment scheduling in container terminals

Operational Cost | **Equipment Capacity** | **Throughput in a yard**

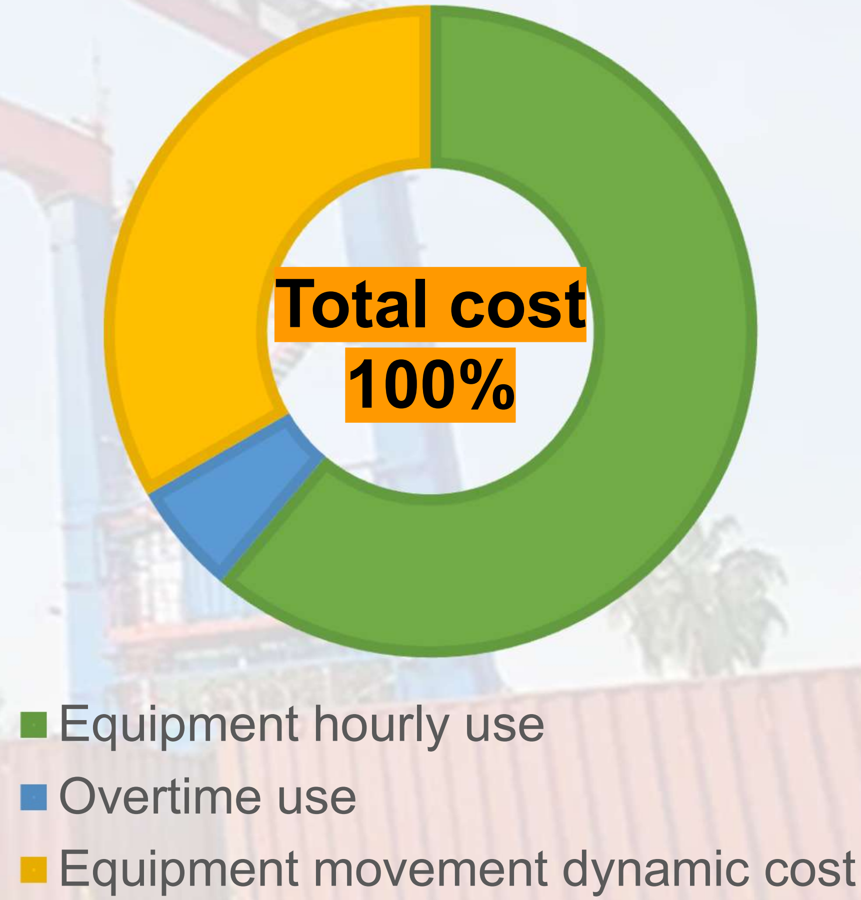
Inputs

Workload	Demand per block per time period
Equipment	Capacity, starting, maintenance schedule
Costs	Equipment usage, overtime cost, dynamic cost for movement

Constraints

- Demand: Fulfil daily demand
- Capacity: Respect equipment capacity
- Shift timing: Maintain working hrs
- Maintenance schedule: Respect maintenance break
- Break schedule: Respect break time if any

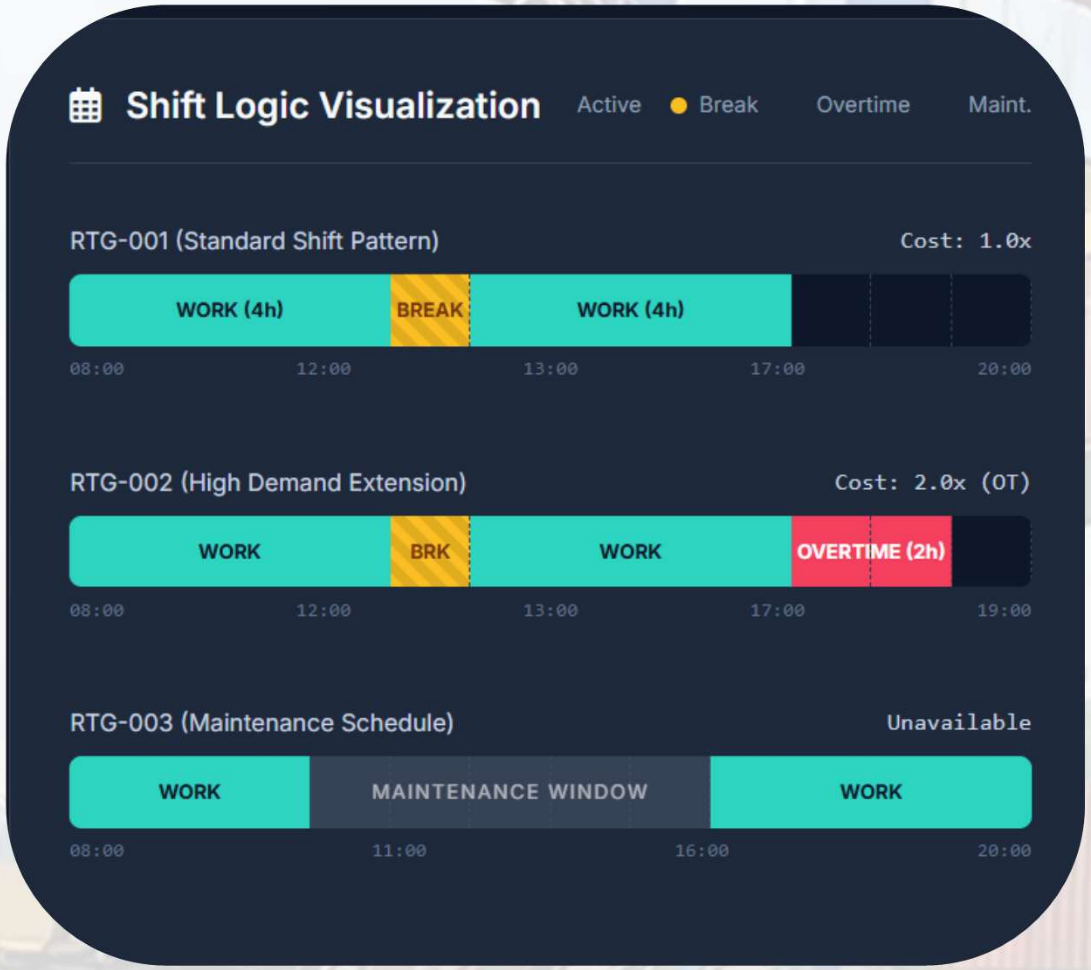
Cost breakdown



Equipment scheduling in container terminals

⬇ Operational Cost | Equipment Capacity | ⬆ Throughput in a yard

Outputs



Business values and impact

- ✓ Cost reduction
- ✓ Better utilization of equipment
- ✓ Scalable solution
- ✓ Data driven decisions

18%

Reduce on total operational costs

12%

Increase in total efficiency in the yard

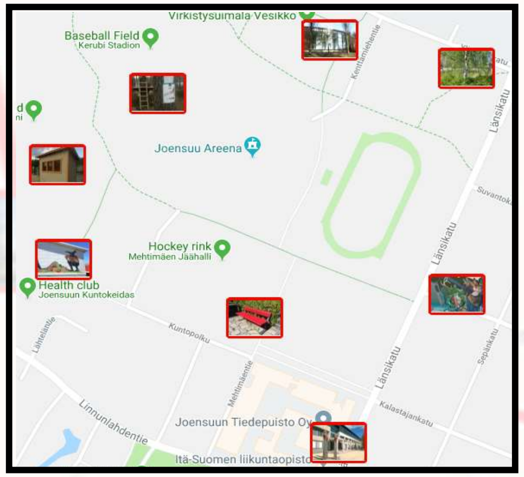
ACADEMIC WORK SAMPLE

SOLVING OPEN-LOOP TSP: IS IT DIFFERENT?

L. Sengupta, R. Mariescu-Istodor, and P. Fränti

Which Local Search Operator Works Best for the Open-Loop TSP?

Applied Sciences. 9(19):3985 (2019)



TSP instance

Closed-loop optimum

Open-loop optimum

TSP instance

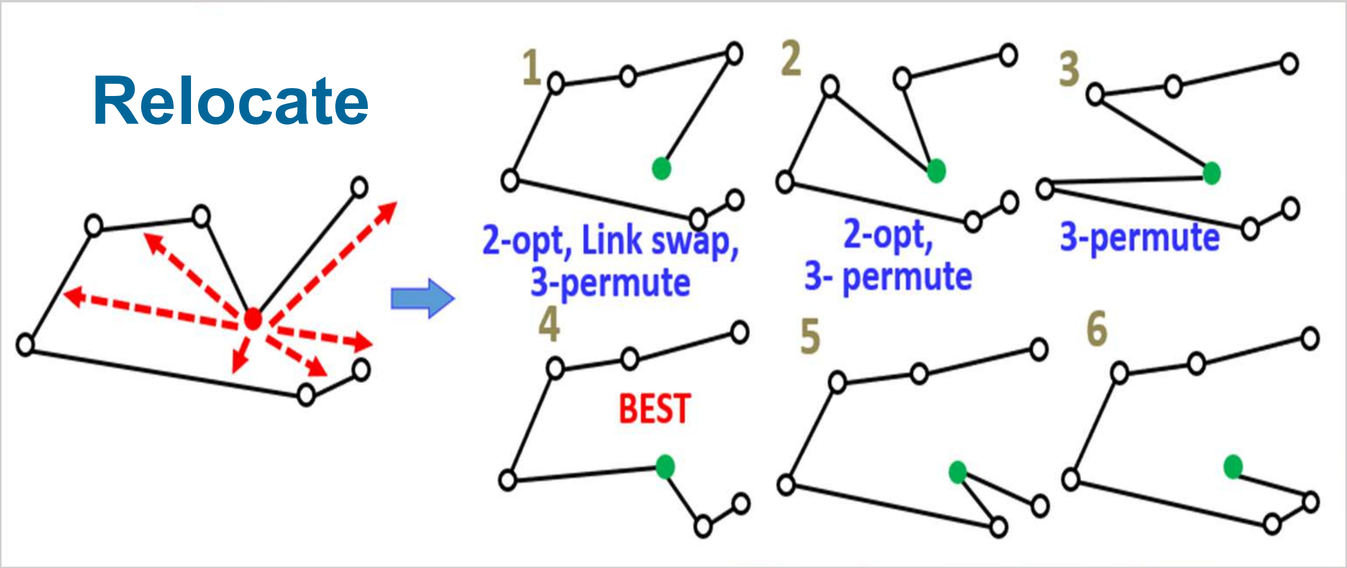
Closed-loop optimum

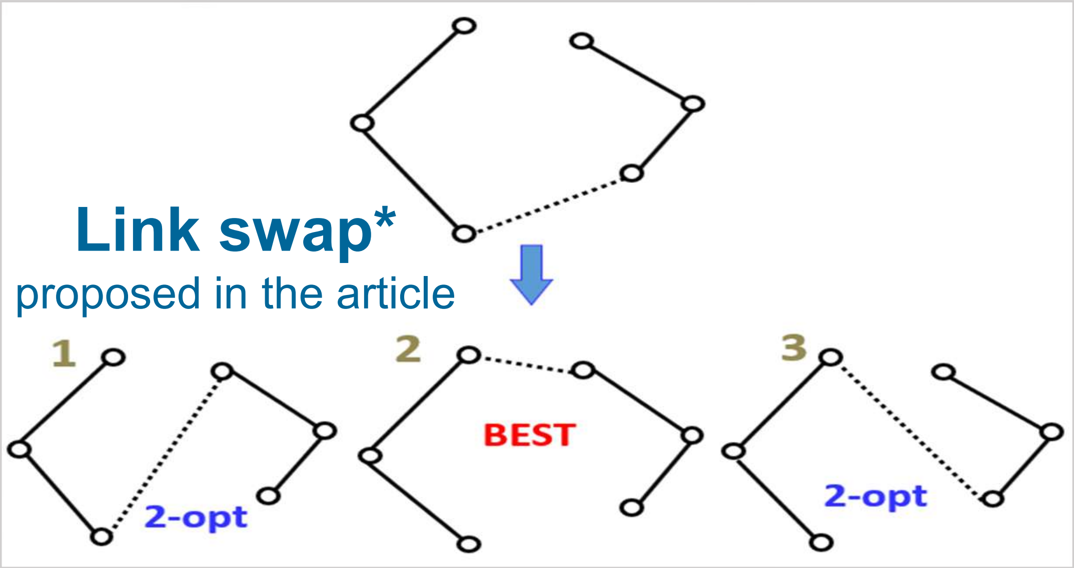
Open-loop sub-optimal

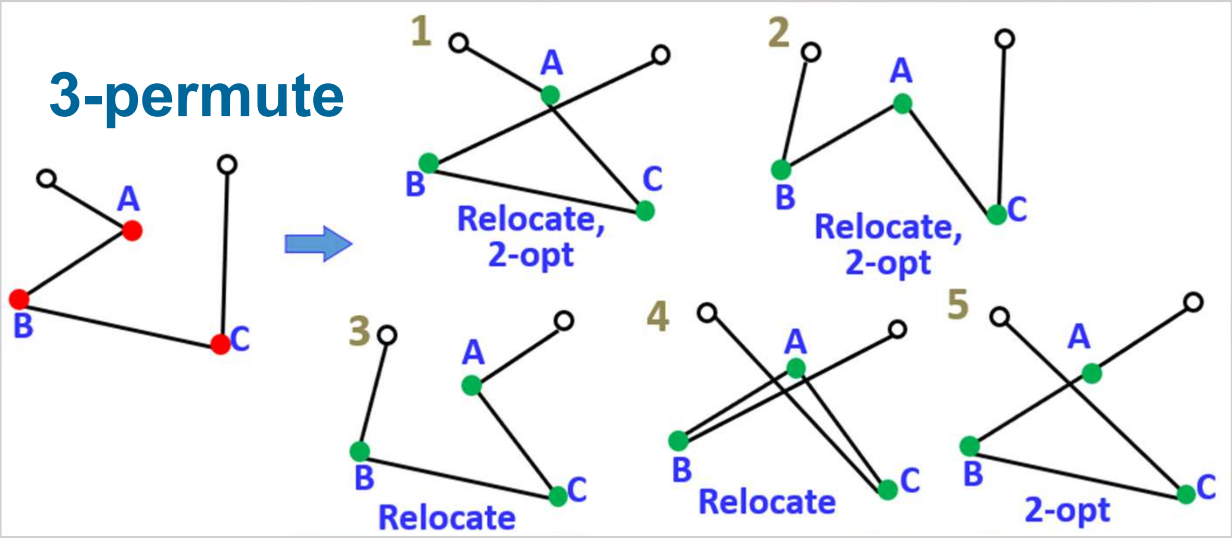
Open-loop optimum

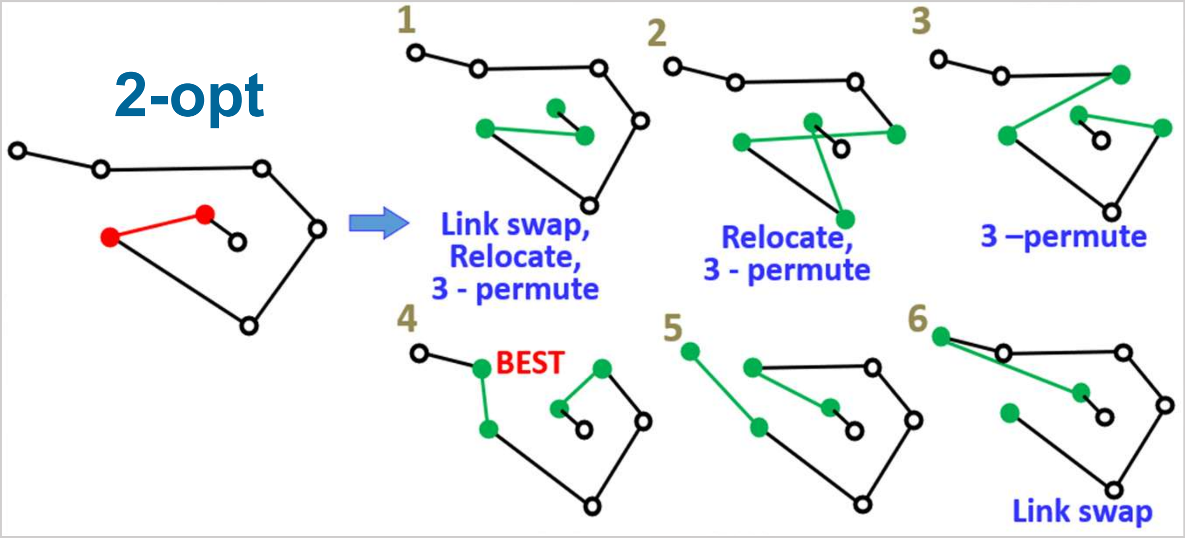
Solving open-loop TSP: Using local search

Local search operators are simple



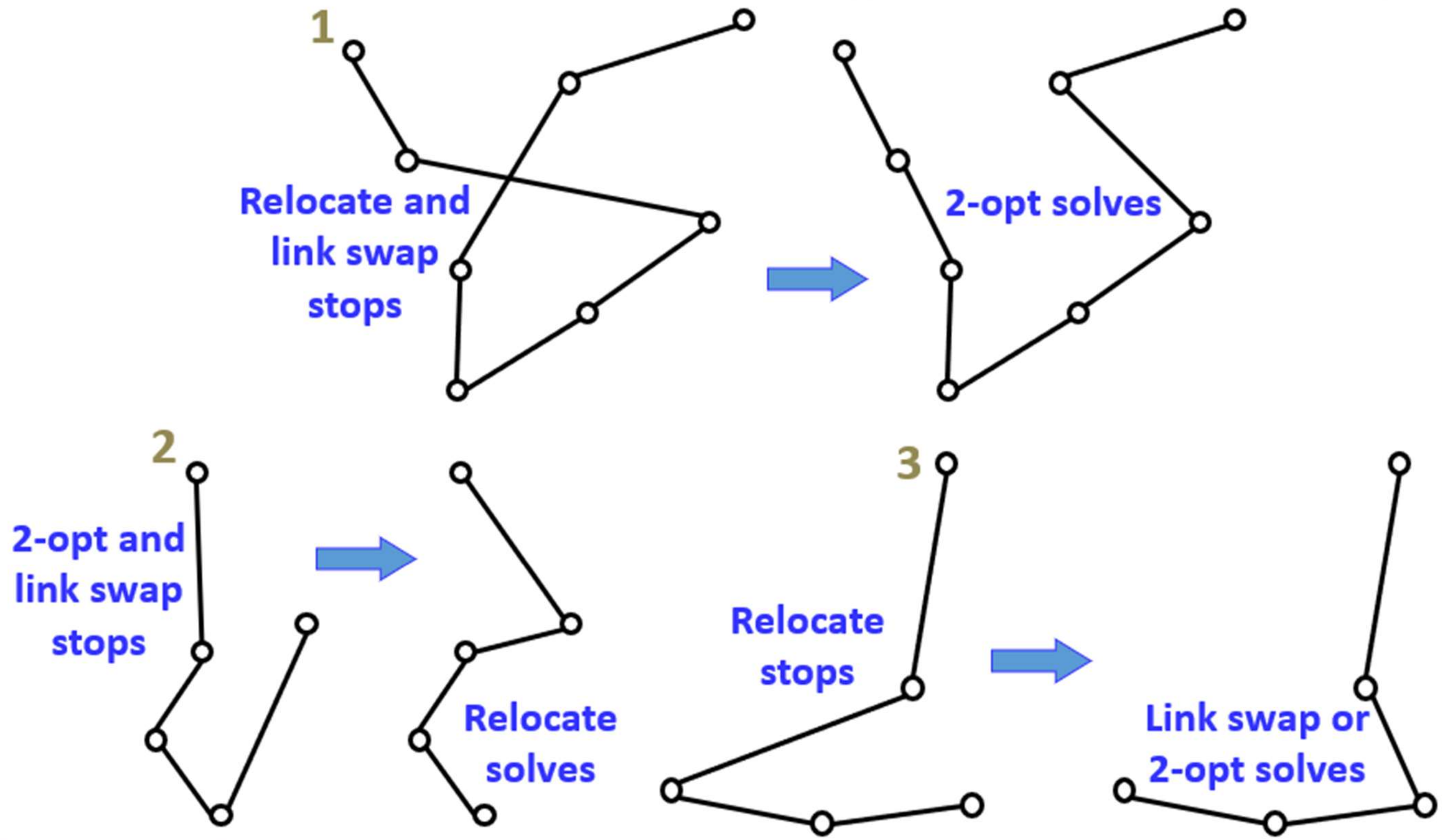






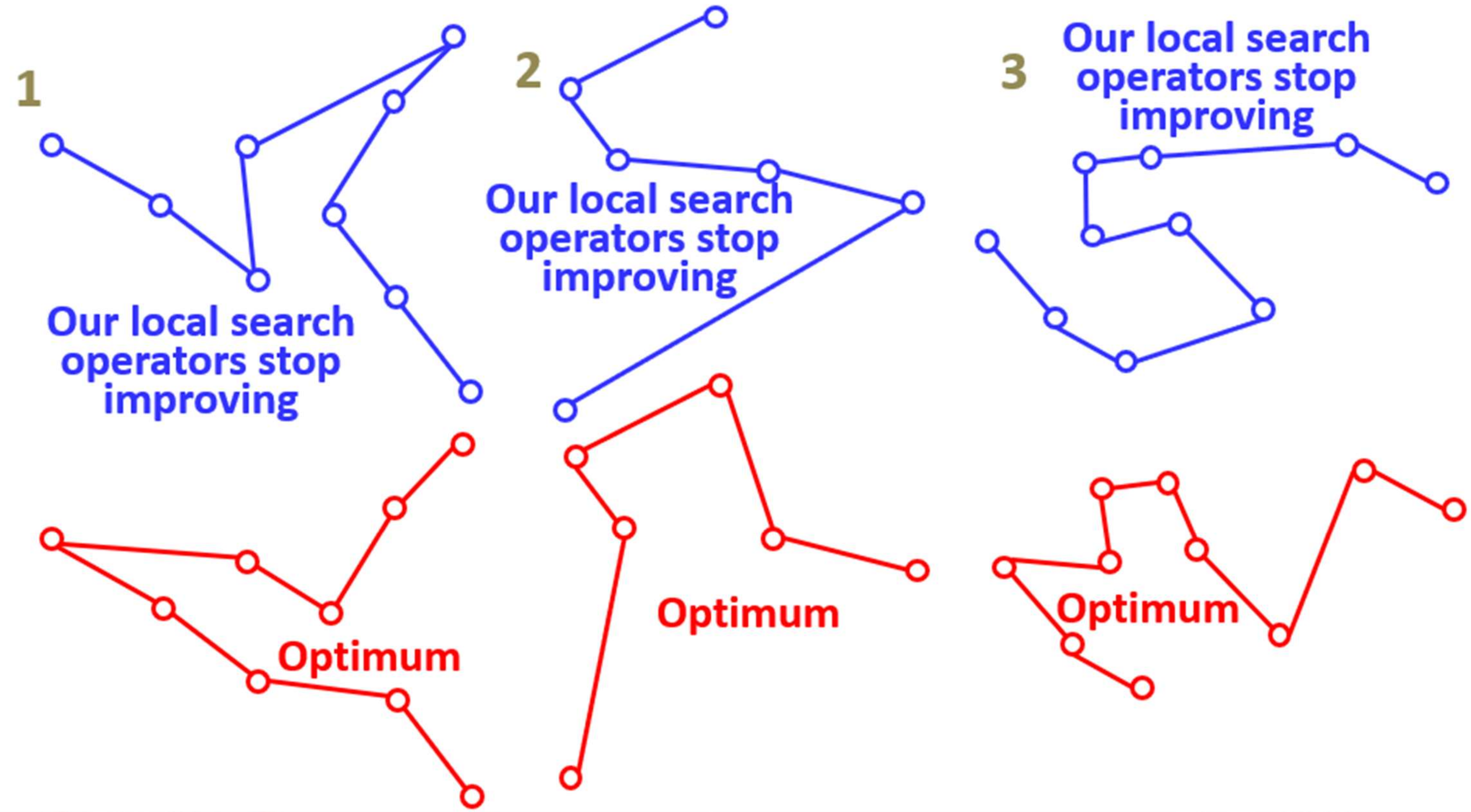
Solving open-loop TSP: Using local search

Local search operators have complementary search-spaces



Solving open-loop TSP: Local optima

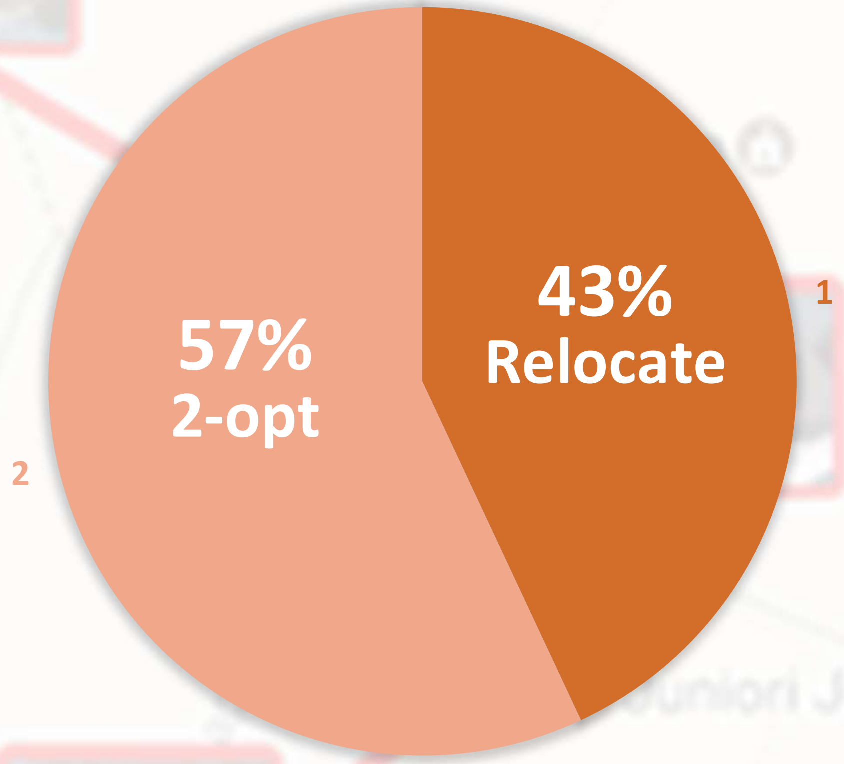
Repeated local search recovers local optima



Solving open-loop TSP: productivity of operators

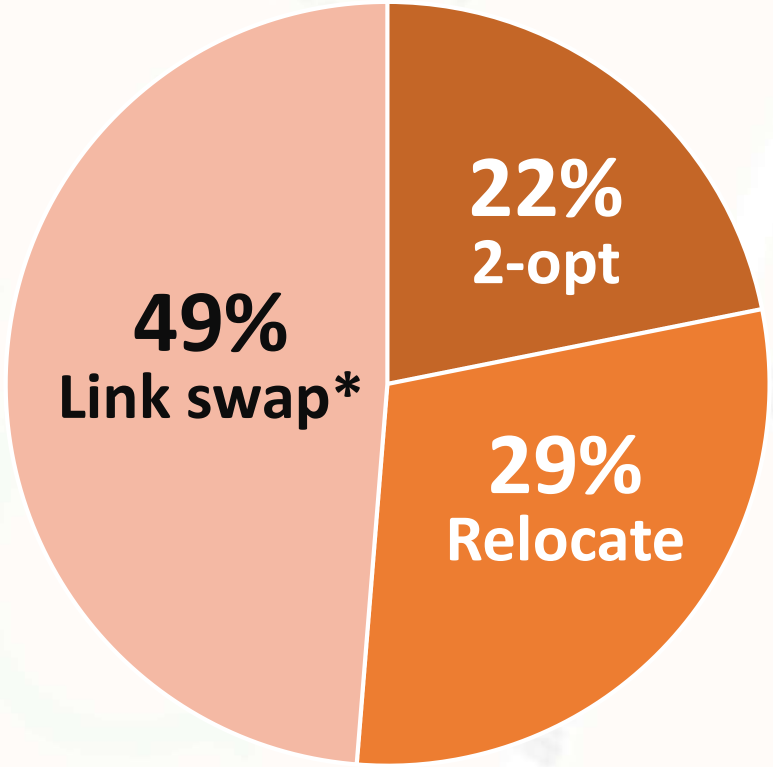
Proposed operator (link swap) is the most efficient

Closed-loop cases



TSPLIB dataset

Open-loop cases



O-Mopsi dataset